

Efficient On-Device Text Embeddings for Personal Knowledge Bases

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Abstract

Personal knowledge management tools increasingly store heterogeneous content – typed notes, imported documents, and screenshots – that users expect to be searchable by meaning rather than exact keyword match. Cloud-based embedding services address this but introduce latency and privacy concerns for personal data. We present a study of on-device text embedding models sized for mobile deployment, evaluating the trade-off between embedding quality and inference latency across three model families. We show that a distilled 22M-parameter model retains 94% of the retrieval accuracy of a server-scale model while running in under 15ms per note on a mid-range mobile SoC, making full on-device "related notes" retrieval practical for consumer note-taking applications.

1. Introduction

As personal note-taking apps accumulate thousands of items over time, users lose the ability to browse for related content manually. Semantic search over embeddings offers a natural solution: rather than requiring exact keyword overlap, notes can be retrieved by meaning. However, most production embedding systems assume a server round-trip, which is a poor fit for personal, sometimes sensitive, notes that many users would prefer never leave the device.

This paper investigates whether embedding models small enough to run comfortably on-device can still produce useful similarity rankings for everyday note collections such as journal entries, meeting notes, and imported reference documents.

2. Method

We evaluate three embedding model sizes (110M, 44M, and 22M parameters) distilled from a common teacher model, measuring both mean average precision on a held-out set of paired "related note" annotations and wall-clock inference latency on representative mobile hardware. All models are quantized to 8-bit weights for deployment.

3. Results and Discussion

The 22M-parameter model achieves 94% of the teacher model's retrieval accuracy while reducing per-note embedding latency by roughly 6x, comfortably enabling embedding generation at note-save time without a noticeable delay to the user. We discuss remaining gaps for long-document (multi-page PDF) embedding, which we leave to future work.

References

- [1] Fictional citation for demonstration purposes only.
- [2] This document was generated as sample content for a MediaPipe sample app.